DOMAIN ZERO

DevSync AI

One Team. Multiple Roles.
One Synchronized Codebase.

AI-POWERED COLLABORATIVE DEVELOPMENT

PROJECT PRESENTATION

Team Members

Ravi Shankar Singh • Vineet Singh • Rishi Raj Gupta

Collaborative workflows, Git integration and intelligent development assistance

01 / PURPOSE

The problem we are solving

Fragmented collaboration

Tasks, code changes, reviews and test results can become disconnected across tools.

Unclear ownership

Teams need clear role-based responsibilities and visible handoffs through development.

Slow feedback loops

Bugs, conflicts and project context are harder to track when information is scattered.

Our vision

A shared development workspace where people, tasks, Git activity and AI assistance work together in one coordinated flow.

02 / SOLUTION

Introducing DevSync AI

DevSync AI connects the development lifecycle across a web dashboard, a VS Code extension and a shared backend.

Web Dashboard

Project overview • Team and role management
Task board • Activity feed
AI insights and notifications

VS Code Extension

Project workspace • Task access
Git-aware workflows
Role assistant • Notifications

Backend Platform

Authentication and permissions
Persistent project data
REST APIs • WebSockets
AI and Git integrations

ONE CONNECTED WORKSPACE

Role-based teamwork

CODER

- Picks up assigned tasks
- Implements features and fixes
- Works with branches and commits
- Uses AI coding assistance

DEBUGGER

- Reviews submitted changes
- Investigates defects
- Identifies likely risks and causes
- Uses AI debugging assistance

TESTER

- Prepares and runs test cases
- Validates completed work
- Reports failures and regressions
- Uses AI test-generation assistance

Administrators manage team membership, roles, projects and task assignments.

04 / WORKFLOW

A structured development pipeline



The backend enforces allowed task-state transitions and role permissions for review, approval, testing and merge.

05 / INTELLIGENT ASSISTANCE

AI-powered capabilities

AI Role Assistant

Context-aware assistance for Coder, Debugger and Tester workflows.

AI Conflict Detector

Highlights potential overlap and conflicts in collaborative changes.

AI Contribution Intelligence

Turns development activity into useful contribution insights.

AI Project Memory

Keeps useful project knowledge accessible across work sessions.

Smart Development Pipeline

Supports smoother task handoffs from implementation through testing.

Gemini Integration

Configured to use Gemini 3.5 Flash-Lite for AI requests.

06 / IMPLEMENTATION

Technology architecture

VS Code Extension

TypeScript

Web Dashboard

React + Tailwind CSS

Backend API

Python + FastAPI

Data Storage

SQLite

Live Updates

WebSockets

Version Control

Git + GitHub API

AI Layer

Gemini API

Code Analysis

Python + Git diff

Modular components keep the extension, dashboard and backend independently testable while sharing project data and workflows.

Collaboration with control

Authentication

Protected API access using authenticated user sessions and token-based requests.

Role permissions

Project membership and action permissions help control who can view, edit, review, approve and merge.

Real-time & history

WebSocket project updates, persistent conversations and notifications support shared awareness.

The platform is designed to keep project activity scoped to authorized team members.

Implementation snapshot

Backend verification

85 tests passed
5 warnings reported

Frontend verification

Production build completed
(as reported by the team)

Extension & integration

Extension compilation and full live integration still require final verification.

Implemented platform areas

- Project and team workflows with role-aware task transitions
- Persistent AI conversations and searchable task workspace
- Task notifications and inbox with read/unread state
- VS Code AI role assistant and notification commands
- WebSocket authentication and project-scoped collaboration

09 / PRODUCT WALKTHROUGH

Suggested live demonstration

1. Sign in and create a project
2. Add teammates and assign Coder, Debugger and Tester roles
3. Create a task and assign ownership
4. Open the project in VS Code and connect the extension
5. Use the AI Role Assistant and review notifications
6. Move the task through the development pipeline

DOMAIN ZERO

Build together.
Stay synchronized.

DevSync AI

Team Members

- Ravi Shankar Singh
- Vineet Singh
- Rishi Raj Gupta

One Team. Multiple Roles.

One synchronized codebase — from task creation to tested, merge-ready work.

THANK YOU